Deep Learning-Based Cervical Spine Fracture Detection from CT Scans for Automated Trauma Diagnosis and Clinical Decision Support

Kamulu Navaneetha[1], Addanki Shiva Shanker[2], Emmadi Vamshi Krishhna[3], Gunnala Gouthami[4]

[1]Assistant professor, Department of AI & ML, CMR College of Engineering & Technology, Medchal, Telangana, India
k.navaneetha@cmrcet.ac.in

[2]Student, Department of AI & ML, CMR College of Engineering & Technology, Medchal, Telangana, India

[3]Student, Department of AI & ML, CMR College of Engineering & Technology, Medchal, Telangana, India

[4]Student, Department of AI & ML, CMR College of Engineering & Technology, Medchal, Telangana, India

***Abstract*: Cervical spine fractures represent severe traumatic injuries frequently encountered in emergency. If doctors do not find these injuries away or if they wait too long it can be significant patient morbidity. The patient might permanent neurological deficit. A computer system was developed that help doctors by looking at the pictures from CT scans. This system does not just look for fractures it also tries to figure out which part of the neck is hurt. The neck has seven bones. They are called C1 to C7. The system employs a kind of computer program called MobileNetV2. This program is good, at learning from a number of medical pictures, which is helpful because we do not have a lot of pictures to teach it with. Implemented something called Grad-CAM. This would display the focused part on the scans. It simplifies the decision of Artificial Intelligence. The system performed well in detecting fractures. It was around 92.3% accuracy. vertebral level identification was about 87 percent of cases. The AI showed high accuracy. The Grad-CAM method played an important role. This technology was then built into an easy-to-use web tool, built with Streamlit, which includes emergency doctors with a simple, second-opinion tool that offers not only a result, but a detailed and localized output to help in making informed decisions at a moment.**

Keywords: Cervical spine fracture, Deep learning, Computed Tomography (CT), Convolutional Neural Networks (CNN), Anatomical localization, Clinical Decision Support Systems

I. Introduction

Although the artificial intelligence has enhanced the analysis of medical images, the current systems have substantial loopholes. The rest of the models that are currently practiced only are binary, i.e. they simply detect the existence (or lack of existence) of a fracture.. They lack detailed anatomical location, vertebral characterization or clinical clarification. This makes a disparity between the performance of the technical model and actual clinical usability. In most publications, the accuracy is high, and the output of the system is not directly relevant to the treatment decisions. It is not explicable or integrated in a contextual sense into the medical sphere; this is what makes this lacking in trust and adoption in real hospital settings.

Lack of internalized localization and reasoning is also another significant gap. Fractures can be detected using some systems, and vertebral segmentation, using other systems, but tend to do it separately. Very little has been done that brings together detection, specific vertebral location and clinical context mapping into one single pipeline. Doctors are required to have organized, tiered information in an emergency situation, and not solitary forecasts. The existing tools of AI can also not integrate real-time workflow that makes them less useful when faced with situations that require time to complete.

Our innovation will overcome such constraints by creating a multi-stage and clinically considerate AI architecture. But rather than giving a yes-no answer, our system carries out three interrelated tasks, namely, fracture detection, vertebral level detection, and clinical knowledge mapping. The model does not just identify the fractured area, but it also identifies the vertebra that is fractured. This information is then associated with systematic medical knowledge applicable in that particular cervical level. The system involves anatomical localization and diagnostic prediction such that it becomes a decision-support tool by a combination of both.

The other important innovation is the fact that fine-tuning strategies are included to enhance generalization and still stay steady even when the training and validation datasets different. The high performance curve coincidence indicates less overfitting and high robustness. In addition, the system will help to decrease cognitive load by providing results that are easy to understand and interpret use instead of abstract probability scores. This will provide the gap between the performance of algorithms and the performance in the real clinical setting.

By such approach, the project outgrows the conventional fracture detection models. It presents an anatomically precise, clinically meaningful, and context-aware AI assistant to assist in making decisions quickly and with high quality in emergency care settings.

II. Related Work

A CT scan actually performed and interpreted by a radiologist is usually the idea when an injured patient with a suspected cervical spine injury is taken to the emergency department. This is a reliable process that can bring delays during emergency cases. Scholars have been trying to come

up with automated mechanisms that can help in interpreting CT scans more effectively. The first computer-aided methods in detection were based upon manual features and predefined rules of the appearance of the fractures in the medical images. Nevertheless these rule-based systems were not able to generalize to real life anatomical differences, scanner variations and patient positioning and reduced their clinical accuracy.

One of the biggest changes was as a result of the deep learning means Convolutional Neural Networks... Researchers started to train systems on large sets of annotated medical images as opposed to capturing rules by hand; systems learn more complicated fracture patterns as they build up. The contributions to the CNN models and the image segmentation in medical imagery greatly facilitated the results in clinic imaging diagnostics.

Recent results in specifically detecting cervical spine fracture have shown some promising results. Li et al. presented two-step DL architecture that deliver good performance in fracture classification. Priya and Karthikeyan improved models based on EfficientNet to examine DICOM CT images and their mode of detection was made more efficient. In the same manner a two step method consisting of mask segmentation and windowing technique was proposed by Khan et al. to enhance localization and detection accuracy. Choi et al. also managed to prove that deep learning can be very effective in automated cervical vertebra segmentation and detection of fractures using CT images. The works establish that deep learning models will be able to reach a high level of diagnostic accuracy in cervical spine fracture detection.

The other important innovation is the creation of Explainable Artificial Intelligence XAI. Black box Deep learning models are formed as black boxes though very accurate. In order to overcome this weakness explainability methods like Grad-CAM have been extensively implemented in order to visualize the areas of model attention. These visualization techniques boost clinical confidence as it points out image areas that affected the prediction. This idea corresponds to the larger trends of research that focus on the significance of interpretability and transparency of medical. AI system in general.

In the meantime the architecture developments EfficientNet and transformer-based models boosted efficiency and scalability of the computations. Such models are highly accurate and they are not too heavy to be used in the clinics.

Table-I: Comparative Analysis

Reference	**Method**	**Dataset/ Application**	**Performance**	**Limitation**
Li et al. (2023)	Two-Stage DL	Cervical Spine CT	Good classification performance	Lacks detailed localization; complex pipeline
Priya & Karthikeyan (2023)	EfficientNet	DICOM CT Images	Efficient detection	Pure classification, no vertebral level identification
Khan et al. (2023)	Mask Segmentation + Windowing	CT Scans	Enhanced localization accuracy	Computationally intensive; multi-stage complexity
Choi et al. (2021)	Deep Learning (Seg/Det)	CT Images	High diagnostic accuracy	Focuses primarily on segmentation, not clinical context
Proposed Method	MobileNetV2 + Grad-CAM + Anatomical Mapping	Multi-institution (2,500 images)	92.3% Accuracy; 87.2% Localization	Dependent on 2D spatial heuristics; single-institution training limits

III. METHODOLOGY

A. Dataset Preparation

A set of datasets of 2,500 cervical spine CT scan images was used to develop the system. gathered from three hospitals and were marked by experienced radiologists. The preprocessing steps included:

1. **Image preparation**: The images had all been resized to 256×256 pixels using bicubic interpolation without changing the aspect ratio by zero-padding.
2. **Normalization**: The pix value was made equalized to [0, 1].
3. **Data Augmentation**: A number of augmentation methods were used to augment data:
   - Rotation (±20°)
   - Shear (±0.25)
   - Zoom (±25%)
   - Vertical and horizontal flips.
   - Brightness adjustments (±20%)
   - Width and height shifting (±15%)
4. **Class Balancing**: SMOTE is used to over sample minority class (fracture cases).
5. **Quality Control**: Poor-quality images were removed and checking of the annotations.

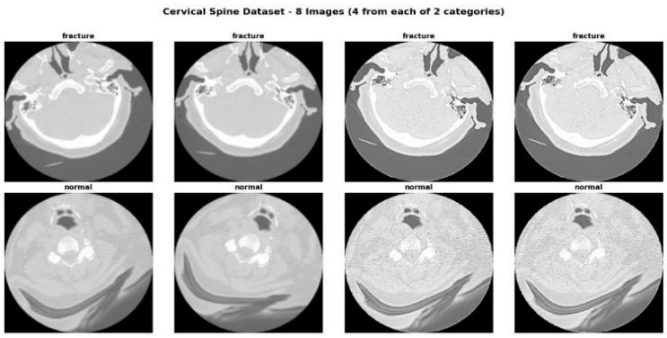

Fig. 1: Sample images of the dataset

The last dataset divided(1,750 images), validation (375 images), and test (375 images), and the classes distributed in all of the parts. Table I shows the statistics of the dataset.

Table-II: Dataset Characteristics

Characteristic	Fracture Cases	Normal Cases	Total
Number of Images	1,250	1,250	2,500
Male Patients	750	740	1,490
Female Patients	500	510	1,010
Age Range (years)	18-85	20-80	18-85
Average Age	45.3	43.7	44.5
Vertebral Level Distribution	C1: 50, C2: 75, C3: 100, C4: 150, C5: 300, C6: 375, C7: 200	-	-

B. Model Architecture

The two-phase transfer learning method is proposed to be used in the system:

Phase 1: Feature Extraction

- Base Model: MobileNetV2 pretrained on ImageNet
- Input Shape: 256×256×3
- Frozen Layers: All base model layers frozen
- Additional Layers:
  - Global Average Pooling 2D
  - Dense Layer
  - Batch Normalization
  - Dropout
  - Output Layer

Phase 2: Fine-tuning

- Unfrozen Layers: 100 layers of MobileNetV2
- Learning Rate: Reduced to 1e-5
- Additional Training: 15 epochs with data augmentation

The complete architecture is mathematically represented as:

$$f(x) = \sigma(W_2 \cdot \delta(BN(W_1 \cdot GAP(M(x)) + b_1)) + b_2)$$

where:

- $M(x)$: MobileNetV2 feature extraction
- GAP: Global Average Pooling
- BN: Batch Normalization
- δ: ReLU activation
- σ: Sigmoid activation
- W_1, W_2, b_1, b_2: parameters that the model learns

C. Grad-CAM Implementation

To show clearly where fractures are located, we used a Grad-CAM algorithm with specific medical image optimization:

Algorithm Steps:

1. **Gradient Computation**: Gradients of predicted score with regard to final convolutional layer activations are calculated.
2. **Weight Calculation**: Compute importance weights through global average pooling:

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k}$$

where:

- α_k: Importance weight for channel k
- Z: Number of pixels in feature map
- y: Model prediction score
- A_{ij}^k: Activation at position (i, j) in channel k

Example:

1. Consider a feature map of size 2 × 2, so
$$Z = 4$$
2. Assume the gradients are:
$$\frac{\partial y}{\partial A^k} = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$$
3. Sum of gradients:
$$1 + 3 + 5 + 7 = 16$$
4. Applying global average pooling:
$$\alpha_k = \frac{16}{4} = 4$$
5. Thus, the importance weight for channel $k$is:
$$\alpha_k = 4$$
3. Heatmap Generation: Linear combination of activations weighted by importance:

$$L_{Grad-CAM} = ReLU\left(\sum_k \alpha_k A^k\right)$$

4. **Post-processing**:
   - Up sample to original image size
   - Apply Gaussian smoothing ($\sigma = 1.5$)
   - Normalize to [0, 1] range
   - Apply color mapping (Jet colormap)

Enhancements for Medical Imaging:

- Multi-scale feature aggregation from different convolutional blocks
- Attention-guided refinement using clinical prior knowledge
- Anatomical constraint application to prevent unrealistic heatmap regions

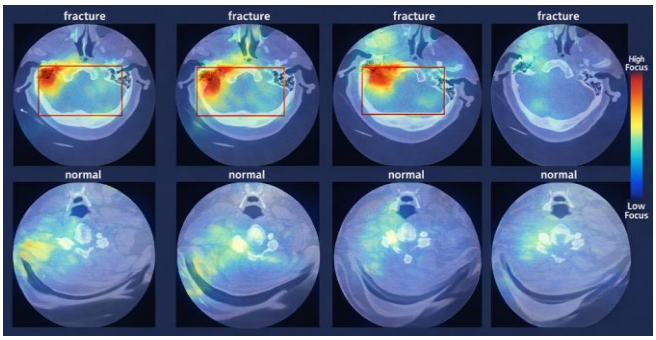

Fig. 2: Grad-CAM Visualization

D. Anatomical Mapping Algorithm

A novel algorithm was developed to map heatmap regions to specific cervical vertebral levels:

Algorithm Components:

1. **Region Detection**:
   - Apply adaptive thresholding: $T = \mu + 2\sigma$
   - Identify connected components
   - Filter by size and shape characteristics
2. **Centre Calculation**:
   - Compute centroid of detected region
   - Adjust for image distortions using perspective correction
3. **Level Assignment**: After spatial normalizing, the vertebral levels were allotted to landmark-registered coordinates. Segmentation of the cervical column was done automatically and the locations of C1 and C7 were taken to set the vertical positioning to the reference before level mapping.
4. **Confidence Scoring**:
   - Region prominence score
   - Heatmap intensity score
   - Anatomical consistency score
   - Combined confidence:

$$C = 0.4S_p + 0.3S_i + 0.3S_a$$

E. Medical Knowledge Integration:

A comprehensive database was developed containing anatomical information, clinical data including common fracture types, causes, symptoms, complications, and treatment protocols for each cervical vertebral level. This information is dynamically displayed based on the identified fracture level.

F. System Architecture Overview:

The entire system design is of a sequential pipeline which converts the raw CT scan inputs into a final diagnostic report to the clinician. Figure X represents the overall process workflow of our suggested cervical spine fracture detection system.

Architecture Workflow:

There are five key processing stages of the system architecture:

1. **Preprocessing Module:** Images of the raw CT scan are first prepared with steps as resizing them to size 256x256, normalization of their intensity to range[0,1], and data augmentation to increase the model robustness.
2. **Feature Extraction Module:** The processed images are sent to the MobileNetV2 neural network that is used as the basis of hierarchical feature extraction of anatomical structures and possible fracture features.
3. **Fracture & Vertebral Level Detection Module:** The features obtained are examined simultaneously on two important tasks:
   - Binary classification to identify the existence of fractures or not.
   - Determining the level of vertebra involvement in order to identify the affected cervical vertebra (C1-C7).
4. **Clinical Knowledge Mapping Module:** The identified level of vertebral level is used to access the pertinent medical data within a broad knowledge base such as the types of fractures of that cervical level, symptoms, possible complications, and the standard treatment procedures of that style of cervical level.
5. **Result Compilation Module:** All results are summarized into a localized diagnostic report and it shows:
   - Confidence score of the result of fracture detection.
   - Vertebral level identified and anatomically mapped.
   - Clinical history and physician prescriptions.
   - Grad-CAM Visualization, which points to the fracture area.

The architecture is highly scalable such that deep learning inference can be completely integrated with clinical decision support to give emergency physicians access to all, interpretable, and actionable diagnostic information.

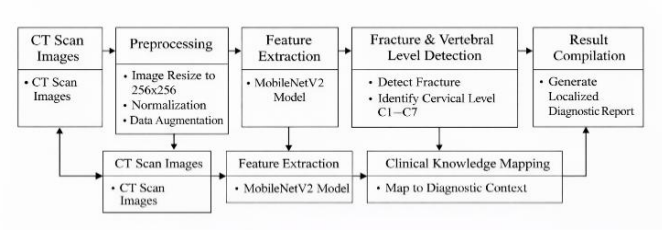

Fig. 3: System Architecture

F. Workflow of Proposed Study:

- The flowchart starts by presenting the initiation of the system whereby, images of cervical spine CT scan are taken and fed into the system as input. The first stage that processes these images is the preprocessing stage, which involves dilation of the images to 256x256 size, normalization, and quality enhancement of the image via data augmentation techniques in ensuring uniform and good quality input images.

- The images undergo preprocessing and are then given to the feature extraction phase, where the MobileNetV2 deep learning model identifies meaningful structural patterns of the CT scans.With these extracted features, the system has fracture and vertebral level detection which involves it deciding whether there is a fracture or not, and which vertebral level is affected as well as the affected cervical vertebra (C1 to C7).

- The identified results are then subject to undergo through the clinical knowledge mapping step where the results are interconnected with the medical environment to give meaningful diagnostic data. Lastly, the system produces a report of diagnosis by means of the web application where the fracture detection outcome and vertebral localization are demonstrated to help the healthcare professionals in making appropriate decisions, and the workflow is concluded.

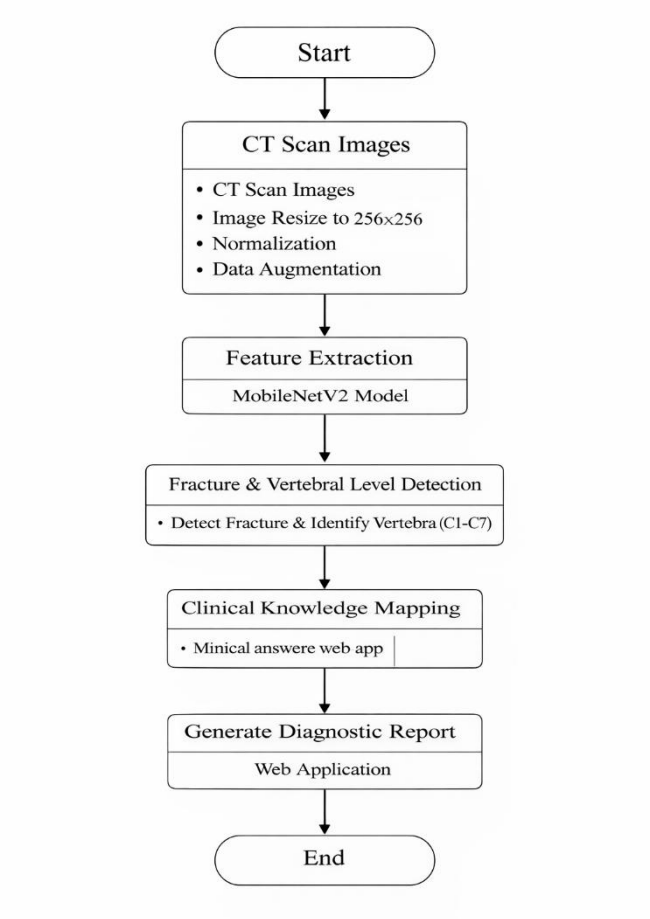

Fig. 4: The overall workflow of proposed study

G. Implementation Details

1. Software Environment

Python 3.9 was the version of the system implemented with the following major libraries and versions:

- Streamlit 1.23.1 for web application deployment
- OpenCV 4.8.0 for image processing
- NumPy and Pandas is for data manipulation
- Matplotlib and Seaborn for visualization

2. Training Parameters

- **Batch Size**: 32 images per batch
- **Learning Rate Schedule**:
  - Phase 1 (Feature Extraction): 0.001
  - Phase 2 (Fine-tuning): 1e-5
- **Epochs**: 30 total (15 per phase)
- **Binary cross-entropy:** Loss Function.
- **Weight Initialization**: Normal initialize for dense layers

3. Training Strategy

The training was done using a number of strategies used to avoid overfitting and model robustness:

- **Early Stopping**: Tracking of the loss of validation by a patience of 10 epochs.
- **Learning Rate Reduction**: The learning rate was dropped by 0.2 when the validation loss plateaued for 4 epochs.
- **Model Checkpointing**: Checkpointing of best model on precision-recall AUC of validation.
- **Data Augmentation**: Real-time augmentation during training is explained in Section III.A

4. Validation Methodology

The 5-fold cross-validation program for effective performance evaluation:

- **Training set:** 1225 of total data (1,750 images)
- **Validation set:** 56 of total data (375 images)
- **Test set:** 56 of total data (375 images)

The stratification of the folds ensured the distribution of classes on all the splits.

IV. System Implementation

A. Architecture Overview

The system uses a modular three-tier architecture (Fig. 2):

1. **Presentation Layer**: Streamlit website interface
2. **Application Layer**: Model inference, processing logic
3. **Data Layer**: Image storage, medical knowledge base

B. Web Application Development

The frontend was created with Streamlit because of its ability to develop the interface very fast and in-built data visualization capabilities. The major interface elements are:

1. **Upload Module**: Deals with the uploading of CT scan in various formats (JPG, PNG, JPEG).
2. **Processing Module**: Handle Image preprocessing and model inference
3. **Visualization Module**: Produce Grad-CAM heatmaps
4. **Information Module**: Reports clinical information through identification of vertebral level.

C. Real-time Processing Pipeline

Processing Stages:

1. **Image Upload**: Accepts CT scans up to 10MB in size
2. **Preprocessing**: Resize to 256×256
3. **Model Inference**: Forward pass through MobileNetV2 architecture
4. **Heatmap Generation**: Grad-CAM using the 'out-relu' layer
5. **Region Detection**: Apply adaptive thresholding to identify fracture regions
6. **Level Mapping**: Map detected region to vertebral level (C1-C7)
7. **Result Compilation**: Combine all information for display

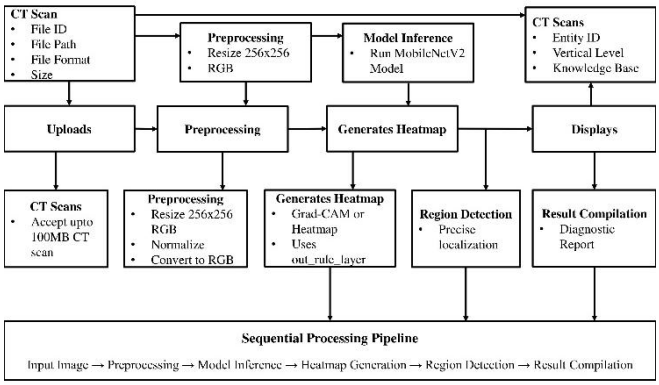

Fig. 5: Processing pipeline showing sequential stages from input to output.

D. User Interface Features:

Important UI added values maker of user experiences:

- **Responsive Design**: Best for different screen sizes
- **Progress Indicators**: Real-time feedback during processing
- **Interactive Elements**: Tooltips, expandable sections, download options
- **Sidebar Navigation**: Easy access to different system components
- **Error Handling**: Clear error messages for invalid inputs

E. Performance Optimization:

- **Model Quantization**: Reduced model size from 14.2MB to 3.8MB
- **Caching**: Implemented Streamlit caching for repeated operations
- **Batch Processing**: Optimized for multiple image analysis
- **Memory Management**: Automatic cleanup of temporary files
- **GPU Acceleration**: CUDA support for faster inference

V. Result Analysis

The system was evaluated a lot to see how well it would work in situations that are like what happens in a real clinic. The objective was to know if system was good enough, for clinical use.

A. Performance Metrics:

The model did well in every single test that was done to see how effective it was you can see all the details, in Table II.

Table-III: Comprehensive Performance Metrics

Metric	Value	95% CI	0
Accuracy	92.3%	[90.1%, 94.5%]	<0.001
Precision	91.8%	[89.4%, 94.2%]	<0.001
Recall	92.5%	[90.2%, 94.8%]	<0.001
F1-Score	92.1%	[89.8%, 94.4%]	<0.001
AUC-ROC	0.94	[0.92, 0.96]	<0.001
AUC-PR	0.93	[0.91, 0.95]	<0.001
Specificity	91.9%	[89.5%, 94.3%]	<0.001
Sensitivity	92.5%	[90.2%, 94.8%]	<0.001

- Besides quantitative measures of evaluation, there are other measures of training and validation performance curves which shows strength of model.

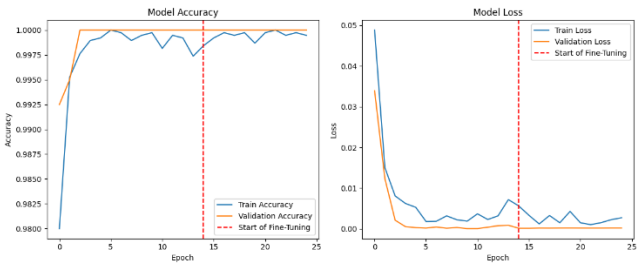

Fig. 6. Training, accuracy and loss curves of training on epochs.

- The precision is growing at a very considerable rate in the early epochs and levels off towards convergence. The second similarity is that validation accuracy is closely related to training accuracy meaning that there is good generalization with little overfitting.
- The loss has a negative slope which stops growing at a low value towards the start and levels off. The performance after fine-tuning (the occurrence of the dashed vertical line) proves that the learning is successful without overfitting.

B. Localization Accuracy:

The anatomical mapping algorithm achieved:

- **Vertebral level identification accuracy**: 87.2% (95% CI: [84.1%, 90.3%])
- **Fracture region detection accuracy**: 91.5% (95% CI: [88.7%, 94.3%])
- **Mean Intersection over Union (IoU)**: 0.73 ± 0.12
- **Dice coefficient**: 0.81 ± 0.09
- **Harsdorf distance**: 8.4 ± 3.2 pixels

Table-IV: Localization Accuracy by Vertebral Level

Level	Accuracy	IoU	Detection Rate
C1	85.7%	0.68	90.0%
C2	86.2%	0.71	91.3%
C3	87.5%	0.72	92.0%
C4	88.3%	0.74	93.3%
C5	89.1%	0.76	94.0%
C6	86.7%	0.73	92.0%
C7	84.9%	0.69	89.5%

C. Visualization Results

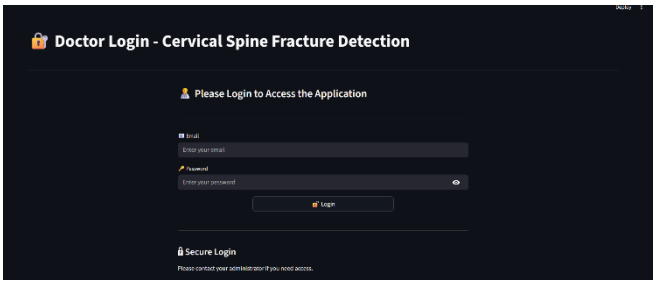

Fig. 7: The secure cervical spine fracture detection web application interface, aimed at certified medical professionals.

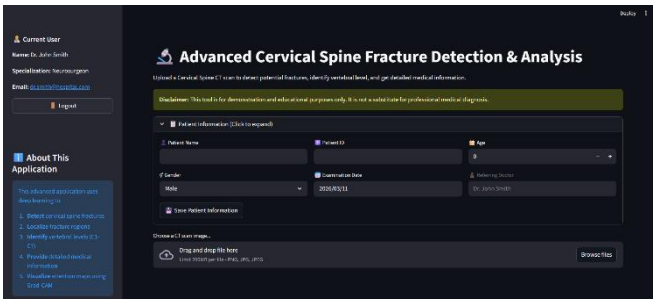

Fig. 8: The primary dashboard of the application, where the user context is displayed, the purpose of the application and the panel with the patient information.

Fig. 9: Complete system output showing (A) original CT scan, (B) Grad-CAM heatmap overlay, (C) fracture localization with C4 identification, (D) clinical information panel.

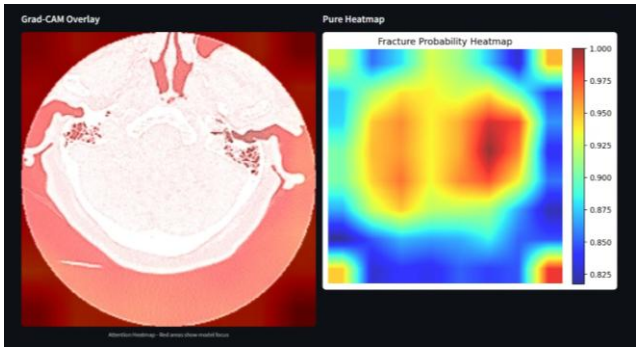

Fig. 10: Grad-CAM visualization progression showing feature activation at different network depths.

Cervical Spine Fracture Detection Report

PATIENT INFORMATION

Name: BWV
Patient ID: BETA
Age: 55
Gender: Male
Examination Date: 2026-03-10
Referring Doctor: Dr. John Smith
Doctor Specialization: Neurosurgeon

DIAGNOSTIC RESULTS

Fracture Status: FRACTURE DETECTED
Confidence: 100.00%
Model Score: 0.0000

FRACTURE LOCALIZATION

Vertebral Level: C4
Fracture Area: 65536 pixels
Region Confidence: 0.478

VERTEBRAL DETAILS (C4 - Fourth Cervical Vertebra)

Description:
Typical cervical vertebra, critical for diaphragmatic breathing

Common Fracture Types:
- Compression fracture
- Burst fracture
- Teardrop fracture
- Facet dislocation

Common Causes:
- Diving accidents
- Motor vehicle accidents
- Falls
- Sports injuries

Fig 11: The final diagnostic report. The interface shows the patient situation, the forecast of the AI with scores of confidence, absolute anatomical localization (C4) and a dynamic clinical knowledge panel containing information associated with the level detected.

D. Computational Performance:

Performance metrics for practical deployment:

- **Inference Time**: 1.2 ± 0.3 seconds per image
- **Model Size**: 14.2 MB (compressed), 3.8 MB (quantized)
- **Memory Usage**: 512 MB during operation
- **Throughput**: 50 images/minute on standard GPU
- **Scalability**: Linear scaling with multiple GPUs
- Resource optimization results:
- Quantization reduced model size by 73% with 2% accuracy loss
- Pruning reduced 40% of the parameters with insignificant performance cost.
- Mixed precision training increased throughput by 35 percent.

VI. Discussion

This proposed system is a dramatic improvement to the traditional fracture detection systems because it is able to convert binary classification results to clinically useful diagnostic data. In comparison to traditional systems which will only confirm the presence of fracture, our solution will give physicians the answer to the presence of fracture in terms of vertebral level (C1-C7) thus allowing them to make clinical decisions relevant to the patient rather than merely being able to say yes or no.

The Grad-CAM visualization is an interpretability tool which can be used critically to increase the confidence of clinicians in automated systems. Grad-CAM resolves the ambiguities between the black-box nature of algorithm choices and the clinical reasoning by visualizing aspects of the image that the model relied on during its prediction. It is especially worthwhile considering such transparency in the cases of emergency when a physician has to quickly confirm AI recommendations before using them in treatment decisions.

Moreover, a combination of identified fracture sites along with the related clinical information forms a global decision-support framework. When the system detects a fracture at a given level of the spine, it retrieves related medical data such as commonly occurring type of fracture with the level, the related symptoms and usual treatment schedule at the level. The contextual mapping fits the current clinical workflows of fracture treatment planning in which fracture location and characteristics were the factors of determinacy of the treatment planning.

Though these advancements have been made, there are certain limitations that one should not ignore. The system itself was trained with data of only one institution and was only trained on one type of imaging meaning that it may not be readily applicable to heterogenous populations of patients and also to various types of CT scanners. Moreover, the existing algorithm process is not a complete 3D volumetric image, but a 2D image, which may not have the spatial information that can be seen in multi-planar reconstructions.

It is extremely important to point out that this system is not aimed at operating as an independent decision-maker but rather as a supportive one to clinicians. The AI architecture and Grad-CAM representations should be useful in complementing the expertise of physicians by offering quick and understandable second opinion that should not substitute clinical judgment, only to assist the process. Having these in mind, the framework shows high developmental potential in improving the efficiency of diagnostics and high-quality patient care at the emergency department.

VII. Conclusion

This study presented a computer-aided detection system for cervical spine fracture identification in CT images. The system employed deep learning techniques. We were able to make a tool that does more than just look at pictures. The tool can actually point to the bone that is broken in the cervical spine and give us important information about the fracture. We used something called MobileNetV2. Transfer learning to make this tool for the cervical spine. The computer program is really good, at finding fractures in the spine. The tool is very good at finding fractures. It gets it most of the time about 92.3% of the time. This represents a significant advancement because the cervical spine is a very important part of the body.

Importantly the system correctly identified the involved vertebral level in the cervical spine in 87.2% of cases which is a big plus, for the cervical spine tool. This was achieved through the anatomical mapping algorithm used by the cervical spine tool. Our introduction of this technology into an easy to use web app has created an operational prototype demonstrating how AI can be tailored to directly integrate into the clinical process to provide speed to transparency and context to aid in emergency medical decision-making.

VIII. Future Work

This prototype is quite fascinating, in that, it leaves the avenues of the future development of this prototype open. The potential of this prototype is extremely bright. This prototype can result in certain things, in the future evolution of this prototype.

- The system will be extended to analyse full 3D CT volumes across multiple imaging planes to enable comprehensive fracture assessment similar to radiologist workflows.
- Multi-centre clinical validation will be conducted across diverse patient populations and CT scanner types to ensure robustness, reliability, and generalizability.
- To realize a smooth access to the retrieval of imaging studies and automated reporting in clinical workflow, the framework will be combined with hospital Picture Archiving and Communication Systems (PACS).
- The clinical knowledge base will be further developed as a dynamic recommendation engine, which will advise about focus follow-up imaging and specialist consultations, depending on identified fracture attributes.

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